

Exposure (Taxonomy)	OR (95% CI)	Effect Size (β)	Standard Error (SE)	P-value	nSNP	Cochran's Q	Q P-value
class Actinobacteria	1.34 (1.08-1.65)	0.2909	0.1072	0.0067	15	30.35	0.0068
family Enterobacteriaceae	1.76 (1.16-2.67)	0.5661	0.2119	0.0076	7	20.99	0.0018
order Enterobacterales	1.76 (1.16-2.67)	0.5661	0.2119	0.0076	7	20.99	0.0018
family Methanobacteriaceae	1.13 (1.03-1.25)	0.1262	0.0492	0.0103	10	4.99	0.8355
order Methanobacteriales	1.13 (1.03-1.25)	0.1262	0.0492	0.0103	10	4.99	0.8355
class Methanobacteria	1.13 (1.03-1.25)	0.1262	0.0492	0.0103	10	4.99	0.8355
phylum Bacteroidetes	0.77 (0.63-0.94)	-0.2614	0.1027	0.0109	10	4.82	0.8500
family Bifidobacteriaceae	1.38 (1.07-1.78)	0.3251	0.1292	0.0119	13	34.93	4.80e-04
order Bifidobacteriales	1.38 (1.07-1.78)	0.3251	0.1292	0.0119	13	34.93	4.80e-04
genus Ruminococcus2	0.84 (0.73-0.96)	-0.1732	0.0695	0.0127	15	10.29	0.7405
class Coriobacteria	1.25 (1.04-1.49)	0.2196	0.0916	0.0165	16	18.61	0.2321
order Coriobacteriales	1.25 (1.04-1.49)	0.2196	0.0916	0.0165	16	18.61	0.2321
family Coriobacteriaceae	1.25 (1.04-1.49)	0.2196	0.0916	0.0165	16	18.61	0.2321
family Acidimicrococcaceae	0.81 (0.67-0.96)	-0.2167	0.0919	0.0184	7	4.66	0.5875
genus FamilyXIAD3011group	0.79 (0.65-0.96)	-0.2363	0.1017	0.0202	13	18.42	0.1036
class Bacteroidia	0.81 (0.67-0.97)	-0.2144	0.0965	0.0262	12	9.01	0.6211
order Bacteroidales	0.81 (0.67-0.97)	-0.2144	0.0965	0.0262	12	9.01	0.6211
genus RuminococcaceaeUCG010	1.26 (1.01-1.58)	0.2339	0.1151	0.0422	6	5.97	0.3087
genus Roseburia	0.80 (0.65-1.00)	-0.2184	0.1106	0.0484	14	20.36	0.0866
genus DefluviitaleaceaeUCG011	1.17 (1.00-1.36)	0.1532	0.0784	0.0507	9	2.24	0.9729
family Peptostreptococcaceae	0.87 (0.75-1.00)	-0.1406	0.0735	0.0557	13	9.91	0.6237
genus LachnospiraceaeFC5020group	0.87 (0.74-1.01)	-0.1439	0.0779	0.0648	13	12.51	0.4053
order MollicutesRF9	1.13 (0.99-1.30)	0.1254	0.069	0.0693	13	10.9	0.5371
class Betaproteobacteria	0.83 (0.68-1.01)	-0.1822	0.1004	0.0695	10	6.15	0.7244
order Desulfotribionales	0.86 (0.73-1.02)	-0.1495	0.0855	0.0806	12	11.98	0.3653
genus Howardella	0.92 (0.85-1.01)	-0.078	0.0454	0.0854	10	8.81	0.4549
genus Intestinibacter	1.13 (0.98-1.30)	0.1218	0.0726	0.0933	14	5.89	0.9502
genus Anaeroflum	0.90 (0.80-1.02)	-0.1028	0.0613	0.0938	10	10.77	0.2916
genus Lachnospira	1.28 (0.96-1.73)	0.2503	0.1508	0.0969	6	5.97	0.3088
genus Erysipelatoclostridium	1.11 (0.98-1.25)	0.1015	0.0626	0.1052	15	13.89	0.4578
genus Sutterella	0.88 (0.75-1.03)	-0.1313	0.0821	0.1096	12	9.26	0.5983
genus Phascolarctobacterium	0.87 (0.73-1.03)	-0.1378	0.0874	0.1149	9	3.92	0.8639
genus Enterorhabdus	0.87 (0.73-1.04)	-0.1387	0.0894	0.1211	6	0.31	0.9975
family Porphyromonadaceae	1.27 (0.94-1.71)	0.2372	0.1542	0.1239	9	13.7	0.0899
phylum Actinobacteria	1.26 (0.94-1.71)	0.235	0.1529	0.1243	15	47.05	1.88e-05
genus Haemophilus	1.11 (0.97-1.27)	0.1046	0.0682	0.1254	9	7.9	0.4433
genus Bifidobacterium	1.21 (0.94-1.54)	0.1882	0.1251	0.1324	13	36.91	2.31e-04
family Ruminococcaceae	1.15 (0.96-1.37)	0.136	0.0909	0.1344	10	8.78	0.4574
genus Collinsella	1.24 (0.93-1.66)	0.2178	0.1463	0.1366	10	18.84	0.0266
genus Olsenella	1.08 (0.98-1.12)	0.0786	0.0529	0.1375	11	13.87	0.1792
order Burkholderiales	0.86 (0.70-1.05)	-0.1548	0.1052	0.1413	10	10.15	0.3384
class Negativicutes	0.85 (0.69-1.06)	-0.1583	0.1086	0.1451	12	14.49	0.2068
order Selenomonadales	0.85 (0.69-1.06)	-0.1583	0.1086	0.1451	12	14.49	0.2068
genus Veillonella	1.15 (0.95-1.40)	0.1427	0.0991	0.1499	6	4.92	0.4255
phylum Firmicutes	0.90 (0.77-1.05)	-0.11	0.0793	0.1651	16	12.05	0.6750
genus LachnospiraceaeUCG004	0.89 (0.75-1.05)	-0.1183	0.0857	0.1672	13	6.54	0.8864
genus LachnospiraceaeUCG008	0.89 (0.75-1.05)	-0.117	0.0849	0.1684	11	19.82	0.0310
genus Ruminococcostorquesgroup	0.85 (0.67-1.08)	-0.1654	0.1218	0.1746	9	8.89	0.3516
genus Butyricimonas	0.88 (0.72-1.06)	-0.1319	0.0976	0.1763	13	19.76	0.0717
genus Bilophila	0.86 (0.69-1.07)	-0.1473	0.1112	0.1853	13	22.0	0.0375
phylum Cyanobacteria	0.91 (0.79-1.05)	-0.0957	0.0728	0.1889	8	5.62	0.5846
family Rikenellaceae	0.89 (0.74-1.06)	-0.1189	0.0918	0.1949	17	22.42	0.1302
phylum Lentisphaerae	1.09 (0.96-1.25)	0.0879	0.068	0.1960	9	11.62	0.1691
class Deltaproteobacteria	0.90 (0.77-1.06)	-0.1032	0.0801	0.1974	13	11.88	0.4550
order Rhodospirillales	1.08 (0.96-1.22)	0.0773	0.0606	0.2020	13	10.08	0.6089
genus Methanobrevibacter	1.09 (0.95-1.26)	0.0897	0.073	0.2187	6	6.14	0.2929
family ClostridialesvadinBB60group	1.08 (0.95-1.23)	0.0783	0.0637	0.2192	15	14.09	0.4430
genus Lachnoclostridium	0.90 (0.75-1.07)	-0.1088	0.0891	0.2223	13	9.98	0.6181
genus Hungatella	0.92 (0.79-1.06)	-0.0881	0.0731	0.2278	5	1.96	0.7435
genus Sellimonas	1.06 (0.96-1.17)	0.0605	0.0504	0.2301	7	1.88	0.9307
order Victivallales	1.09 (0.94-1.27)	0.0905	0.0776	0.2435	8	11.61	0.1141
class Lentisphaeria	1.09 (0.94-1.27)	0.0905	0.0776	0.2435	8	11.61	0.1141
class Melainabacteria	0.93 (0.83-1.05)	-0.0706	0.0608	0.2459	10	4.76	0.8549
order Bacillales	0.93 (0.83-1.05)	-0.0679	0.0607	0.2635	8	10.45	0.1645
family Desulfotribionaceae	0.91 (0.77-1.08)	-0.0959	0.0869	0.2696	10	8.47	0.4872
genus Eubacteriumrectalegroup	0.89 (0.72-1.10)	-0.1155	0.1064	0.2777	9	7.71	0.4623
genus Blautia	0.89 (0.72-1.10)	-0.115	0.1064	0.2798	13	15.63	0.2086
genus Eubacteriumoxidoreducensgroup	1.09 (0.93-1.28)	0.0856	0.0809	0.2900	5	2.01	0.7341
genus Butyricococcus	1.18 (0.87-1.61)	0.1654	0.1579	0.2950	8	17.29	0.0156
genus Slackia	1.14 (0.89-1.47)	0.1332	0.1283	0.2991	6	11.94	0.0356
family Lactobacillaceae	1.14 (0.88-1.47)	0.129	0.1294	0.3187	8	26.9	3.47e-04
genus Anaerotruncus	1.10 (0.91-1.33)	0.0948	0.096	0.3231	12	12.18	0.3500
genus ChristensenellaceaeR7group	0.87 (0.66-1.15)	-0.1377	0.1411	0.3293	9	12.84	0.1174
genus Victivallis	1.05 (0.95-1.16)	0.0497	0.0511	0.3312	11	12.97	0.2255
genus Oscillospira	0.92 (0.78-1.09)	-0.0836	0.0871	0.3369	8	5.67	0.5789
genus Escherichia Shigella	1.09 (0.91-1.30)	0.086	0.091	0.3449	10	3.65	0.9330
genus Anaerostipes	1.09 (0.91-1.31)	0.0867	0.0919	0.3455	13	11.21	0.5109
genus Actinomyces	1.07 (0.92-1.25)	0.0706	0.0772	0.3603	7	4.76	0.5756
genus Peptococcus	1.05 (0.96-1.13)	0.0456	0.0501	0.3635	13	11.13	0.5182
order Actinomycetales	1.09 (0.91-1.30)	0.0827	0.0912	0.3646	5	2.77	0.5970
family Actinomycetaceae	1.09 (0.91-1.30)	0.0826	0.0912	0.3650	5	2.77	0.5968
genus Dorea	0.90 (0.71-1.14)	-0.1094	0.1209	0.3656	10	11.7	0.2308
genus Allisonella	0.95 (0.86-1.06)	-0.0484	0.0537	0.3671	8	8.57	0.2846
genus Oscillibacter	0.94 (0.81-1.08)	-0.0643	0.0728	0.3771	13	16.03	0.1900
order Gastranaerophilales	0.95 (0.83-1.07)	-0.0563	0.0638	0.3778	9	5.71	0.6800
family FamilyXIII	1.09 (0.90-1.31)	0.0827	0.0939	0.3785	10	7.11	0.6260
order Clostridiales	0.92 (0.76-1.11)	-0.0834	0.0947	0.3786	12	9.13	0.6097
genus Desulfotribrio	0.92 (0.76-1.11)	-0.0844	0.0984	0.3914	10	14.81	0.0963
genus Veillonellaceae	1.06 (0.93-1.20)	0.0555	0.0648	0.3916	18	16.75	0.4715
genus Turicibacter	0.94 (0.82-1.08)	-0.0578	0.0693	0.4043	11	6.82	0.7423
genus Oxalobacter	0.96 (0.87-1.06)	-0.0429	0.0516	0.4060	11	11.17	0.3448
genus Hoidemanella	1.05 (0.93-1.18)	0.0487	0.0609	0.4235	15	13.27	0.5053
genus Coprococcus2	0.91 (0.72-1.15)	-0.0919	0.1182	0.4368	8	10.55	0.1596
order Pasteurellales	1.05 (0.94-1.17)	0.0442	0.0568	0.4371	14	12.49	0.4882
family Pasteurellaceae	1.05 (0.94-1.17)	0.0442	0.0568	0.4371	14	12.49	0.4882
genus Clostridiumsensustrictum1	1.08 (0.89-1.30)	0.0731	0.0949	0.4414	7	3.51	0.7425
genus Faecalibacterium	1.06 (0.91-1.24)	0.0611	0.0804	0.4471	10	4.63	0.8651
genus Parabacteroides	0.88 (0.63-1.23)	-0.125	0.1711	0.4651	5	6.17	0.1869
phylum Euryarchaeota	1.03 (0.94-1.13)	0.034	0.0466	0.4659	11	10.34	0.4109
family Defluviitaleaceae	1.06 (0.90-1.26)	0.0626	0.0869	0.4711	11	15.36	0.1193
genus Subdoligranulum	0.93 (0.78-1.14)	-0.0701	0.1015	0.4900	11	12.71	0.2404
family Erysipelotrichaceae	1.08 (0.87-1.35)	0.0767	0.1123	0.4947	13	16.75	0.1592
class Erysipelotrichia	1.08 (0.87-1.35)	0.0767	0.1123	0.4947	13	16.75	0.1592
order Erysipelotrichales	1.08 (0.87-1.35)	0.0767	0.1123	0.4947	13	16.75	0.1592
order Lactobacillales	1.07 (0.88-1.28)	0.0638	0.0954	0.5034	15	19.26	0.1552
family Peptococcaceae	1.05 (0.91-1.22)	0.0483	0.0751	0.5201	10	9.11	0.4276
family Christensenellaceae	0.95 (0.82-1.10)	-0.0484	0.0753	0.5209	11	5.59	0.8488
family Rhodospirillaceae	1.04 (0.92-1.17)	0.0377	0.059	0.5226	14	5.96	0.9474
genus RuminococcaceaeUCG014	1.05 (0.90-1.23)	0.0483	0.08	0.5457	11	2.99	0.9815
class Verrucomicrobiae	0.95 (0.82-1.11)	-0.0479	0.0793	0.5461	11	6.3	0.7896
order Verrucomicrobiales	0.95 (0.82-1.11)	-0.0479	0.0793	0.5461	11	6.3	0.7896
family Verrucomicrobiaceae	0.95 (0.82-1.11)	-0.0479	0.0794	0.5465	11	6.3	0.7896
genus Akkermansia	0.95 (0.82-1.11)	-0.0478	0.0793	0.5470	11	6.3	0.7895
family Streptococcaceae	0.94 (0.77-1.15)	-0.0601	0.1005	0.5499	15	21.37	0.0926
genus Fusicatenibacter	1.05 (0.90-1.22)	0.046	0.0779	0.5551	18	9.85	0.9097
genus Butyrivibrio	1.02 (0.94-1.11)	0.0239	0.0419	0.5691	15	18.02	0.2061
genus Marvinbryantia	0.95 (0.80-1.13)	-0.0494	0.0884	0.5764	10	8.05	0.5291
genus Eubacteriumxylophilumgroup	1.07 (0.83-1.39)	0.0708	0.1318	0.5912	9	18.4	0.0184
genus Eubacteriumventriosumgroup	0.94 (0.73-1.21)	-0.0657	0.1298	0.6126	14	32.94	0.0017
genus RikenellaceaeRC9gutgroup	0.98 (0.89-1.07)	-0.0231	0.0465	0.6191	10	4.02	0.9100
family FamilyXI	1.03 (0.93-1.13)	0.0252	0.0512	0.6224	8	7.74	0.3564
family Prevotellaceae	0.95 (0.78-1.16)	-0.0497	0.101	0.6225	16	28.07	0.0211
genus Lactococcus	0.97 (0.88-1.08)	-0.0265	0.0543	0.6257	9	5.15	0.7418
genus Eubacteriumnodatumgroup	0.98 (0.90-1.07)	-0.0207	0.0442	0.6394	11	6.0	0.8151
class Alphaproteobacteria	1.04 (0.88-1.24)	0.0411	0.088	0.6407	7	2.72	0.8431
genus Paraprevotella	0.97 (0.86-1.10)	-0.0282	0.0613	0.6452	13	15.35	0.2230
genus Adlercreutzia	0.96 (0.82-1.13)	-0.0376	0.0821	0.6472	8	5.69	0.5767
genus RuminococcaceaeUCG002	1.03 (0.89-1.19)	0.033	0.0739	0.6557	21	26.95	0.1367
family Oxalobacteraceae	0.98 (0.90-1.07)	-0.0205	0.046	0.6558	14	9.87	0.7047
genus Clostridiuminnocuumgroup	0.98 (0.88-1						